Workshop 11: K-means clustering

The purpose of this exercise is to help you to understand the k-means algorithm. You need to programme the algorithm introduced in the lecture so that you can understand how k-means algorithm works at each step. K-means is a hard boundary algorithm where each data point could only be assigned to one of the clusters. K-means algorithm converges to local optimum and therefore when you start from different centroids it will end up at different cluster results. In this exercise we will understand these points.

Task:

1) Download the file kmeans.csv from blackboard. There are two attributes in the data.

2) Show the scatter plots of the data; you will see that there are three clusters.

3) Write a Python programme to implement the k-means algorithm on the data. Use **Euclidean measure** to calculate the distances. (You can also implement the built-in k-means algorithm in sklearn as a baseline to compare to yours.)

4) Visualise the scatter plot clusters with different colours and their centroids.

5) Visualise the scatter plot clusters with different colours and their centroids at each iteration step.

6) Find the optimal number of clusters. (Find the method described in the lecture.)